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REG NO: 192473007

EXP: 31

DATE:19/06/2026

Aim

To write a Python program to create a **stacked bar plot with error bars** showing the scores of men and women in different groups.

Algorithm

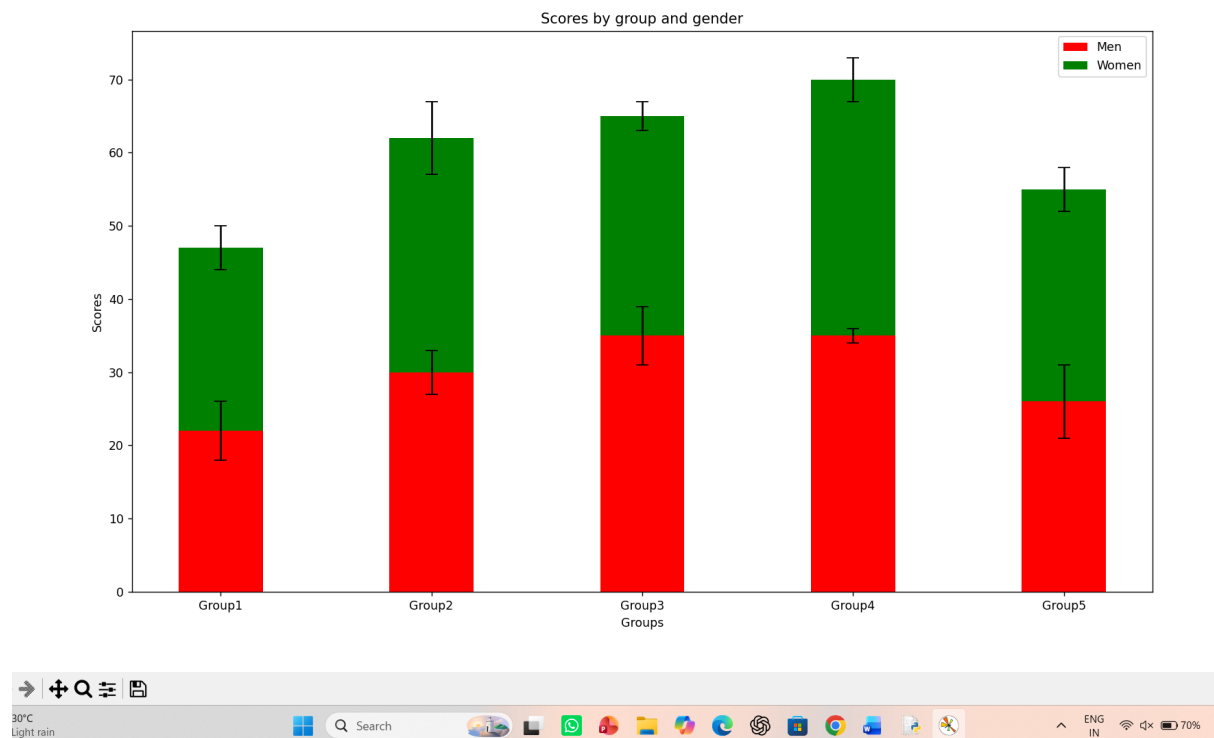
1. Import the required libraries: NumPy and Matplotlib.
 2. Define the mean scores of men and women.
 3. Define the standard deviation values for men and women.
 4. Create positions for the groups using NumPy.
 5. Plot the men's scores as a bar chart with error bars.
 6. Plot the women's scores on top of the men's bars using the bottom parameter.
 7. Add labels for x-axis, y-axis, title, and legend.
 8. Display the stacked bar plot.
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Code

```
import numpy as np
import matplotlib.pyplot as plt
men_means = (22, 30, 35, 35, 26)
women_means = (25, 32, 30, 35, 29)
men_std = (4, 3, 4, 1, 5)
women_std = (3, 5, 2, 3, 3)
groups = ['Group1', 'Group2', 'Group3', 'Group4', 'Group5']
x = np.arange(len(groups))
plt.bar(x, men_means,
        color='red',
        width=0.4,
        yerr=men_std,
        capsize=5,
        label='Men')
plt.bar(x, women_means,
        color='green',
```

```
width=0.4,  
bottom=men_means,  
yerr=women_std,  
capsize=5,  
label='Women')  
plt.xlabel('Groups')  
plt.ylabel('Scores')  
plt.title('Scores by group and gender')  
plt.xticks(x, groups)  
plt.legend()  
plt.show()
```

Output



Result

Thus, the Python program to create a **stacked bar plot with error bars** using Matplotlib was successfully implemented and the required graph was displayed.

EXP:32

DATE:19/06/2026

Aim

To write a Python program to draw a **scatter graph** using random distributions of X and Y values and plot them against each other.

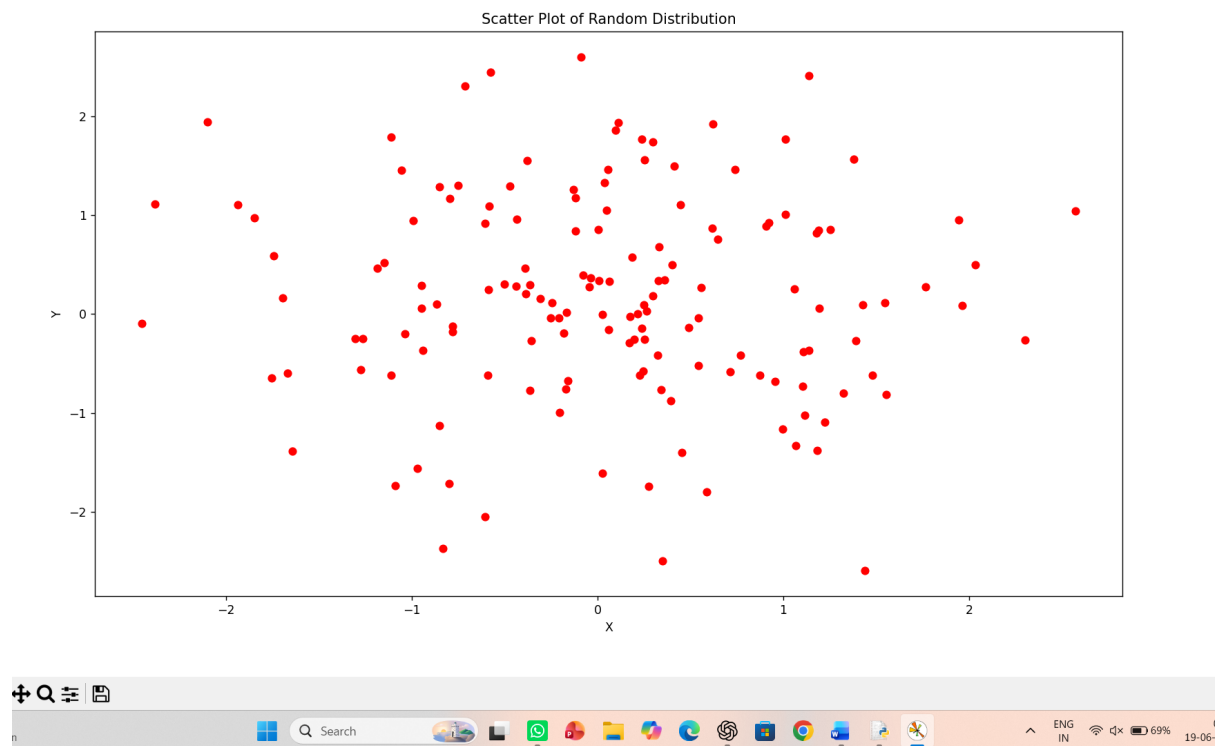
Algorithm

1. Import the required libraries: NumPy and Matplotlib.
 2. Generate random values for X coordinates using NumPy.
 3. Generate random values for Y coordinates using NumPy.
 4. Create a scatter plot using the generated X and Y values.
 5. Set labels for the X-axis and Y-axis.
 6. Display the scatter graph.
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Code

```
import numpy as np
import matplotlib.pyplot as plt
x = np.random.randn(150)
y = np.random.randn(150)
plt.scatter(x, y, color='red')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot of Random Distribution')
plt.show()
```

Output



Result

Thus, the Python program to draw a **scatter graph using random distributions of X and Y values** was successfully executed and the required graph was obtained.

Experiment No: 33

DATE:19/06/2026

Aim

To write a Python program to draw a **scatter plot with empty circles** using random distributions of X and Y values plotted against each other.

Algorithm

1. Import the required libraries NumPy and Matplotlib.
 2. Generate random values for X coordinates.
 3. Generate random values for Y coordinates.
 4. Create a scatter plot using empty circle markers.
 5. Set the edge color of the circles to green and keep the face color empty.
 6. Add labels for X-axis and Y-axis.
 7. Display the scatter plot.
-

Code

```
import numpy as np
import matplotlib.pyplot as plt

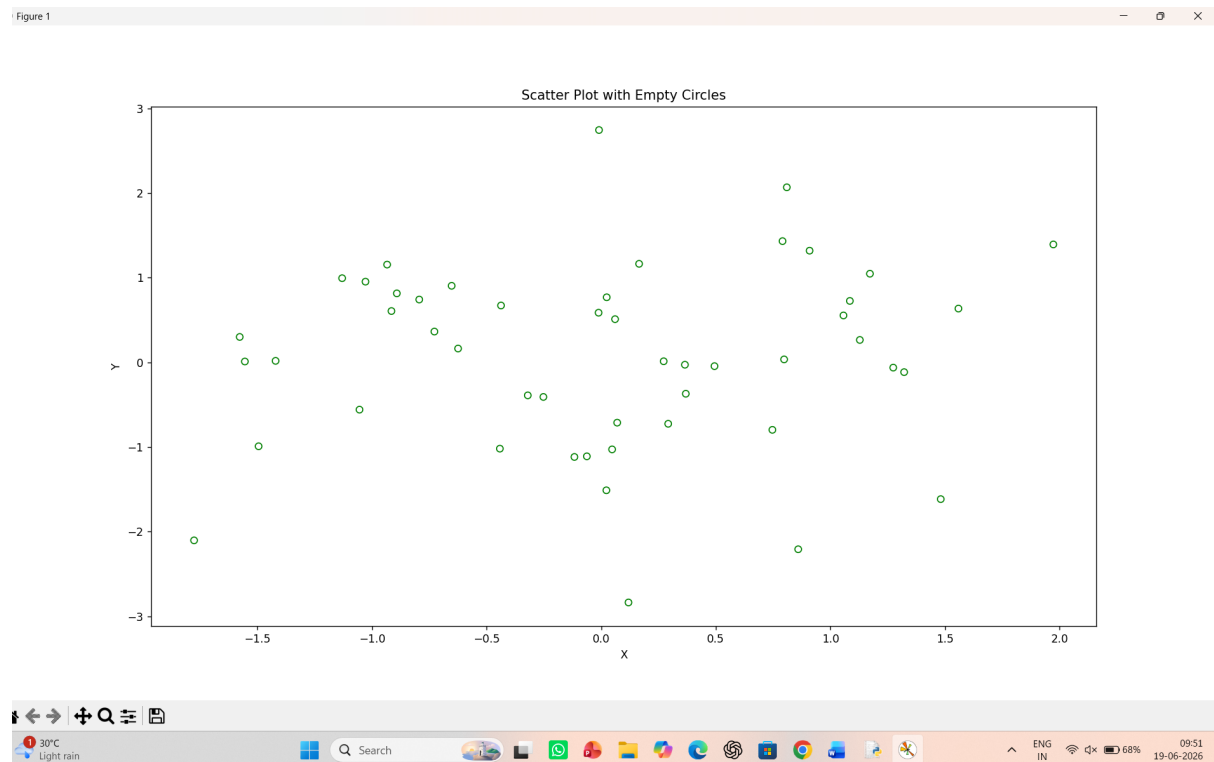
x = np.random.randn(50)
y = np.random.randn(50)

plt.scatter(x, y,
            facecolors='none',
            edgecolors='green')

plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot with Empty Circles')

plt.show()
```

Output



Result

Thus, the Python program to draw a **scatter plot with empty circles using random distributions of X and Y values** was successfully executed and the required graph was obtained.

Experiment No: 34

DATE:22/06/2026

Aim

To write a Python program to draw a **scatter plot using random distributions** and generate balls of different sizes.

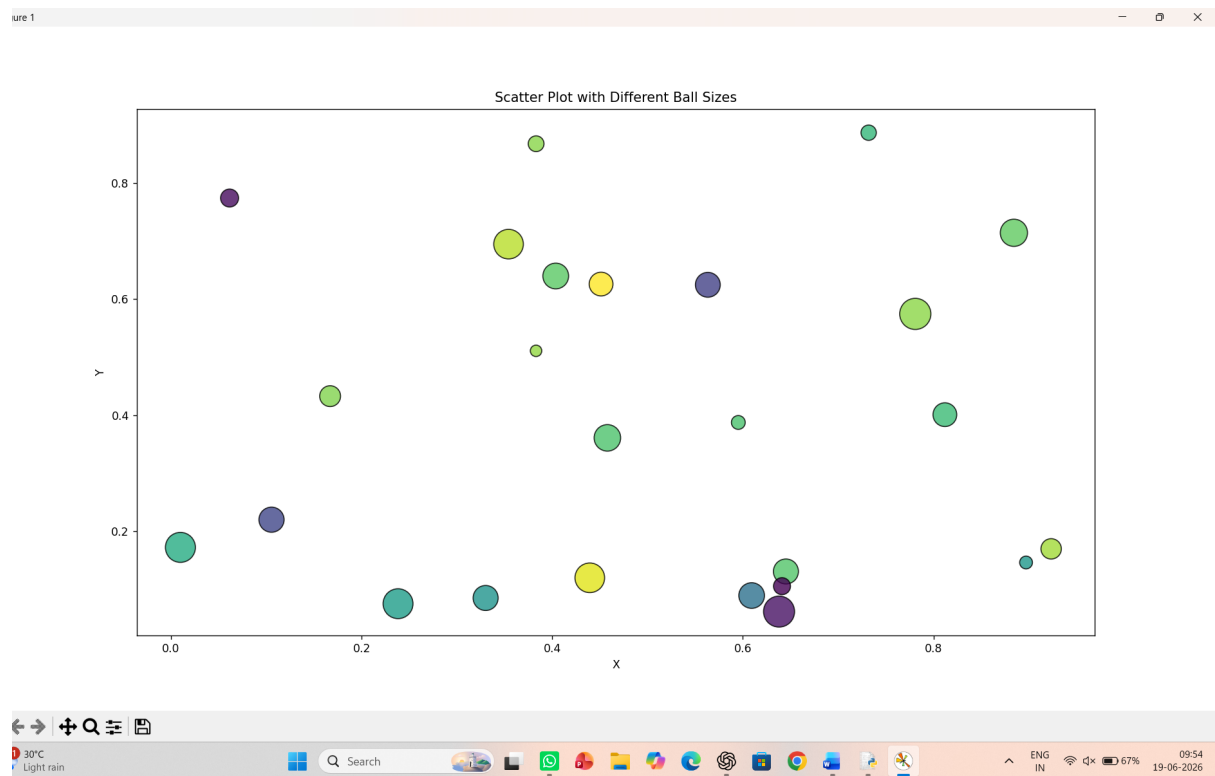
Algorithm

1. Import the required libraries NumPy and Matplotlib.
2. Generate random values for X coordinates.
3. Generate random values for Y coordinates.
4. Generate random values for ball sizes.
5. Generate random colors for the balls.
6. Create a scatter plot using the generated data.
7. Set labels for the X-axis and Y-axis.
8. Display the scatter plot.

Code

```
import numpy as np
import matplotlib.pyplot as plt
x = np.random.rand(25)
y = np.random.rand(25)
sizes = np.random.randint(50, 800, 25)
colors = np.random.rand(25)
plt.scatter(x, y,
            s=sizes,
            c=colors,
            alpha=0.8,
            edgecolors='black')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot with Different Ball Sizes')
plt.show()
```

Output



Result

Thus, the Python program to draw a **scatter plot using random distributions and generate balls of different sizes** was successfully executed and the required graph was obtained.

Experiment No: 35

DATE:22/06/2026

Aim

To write a Python program to draw a **scatter plot comparing Mathematics and Science marks** of 10 students.

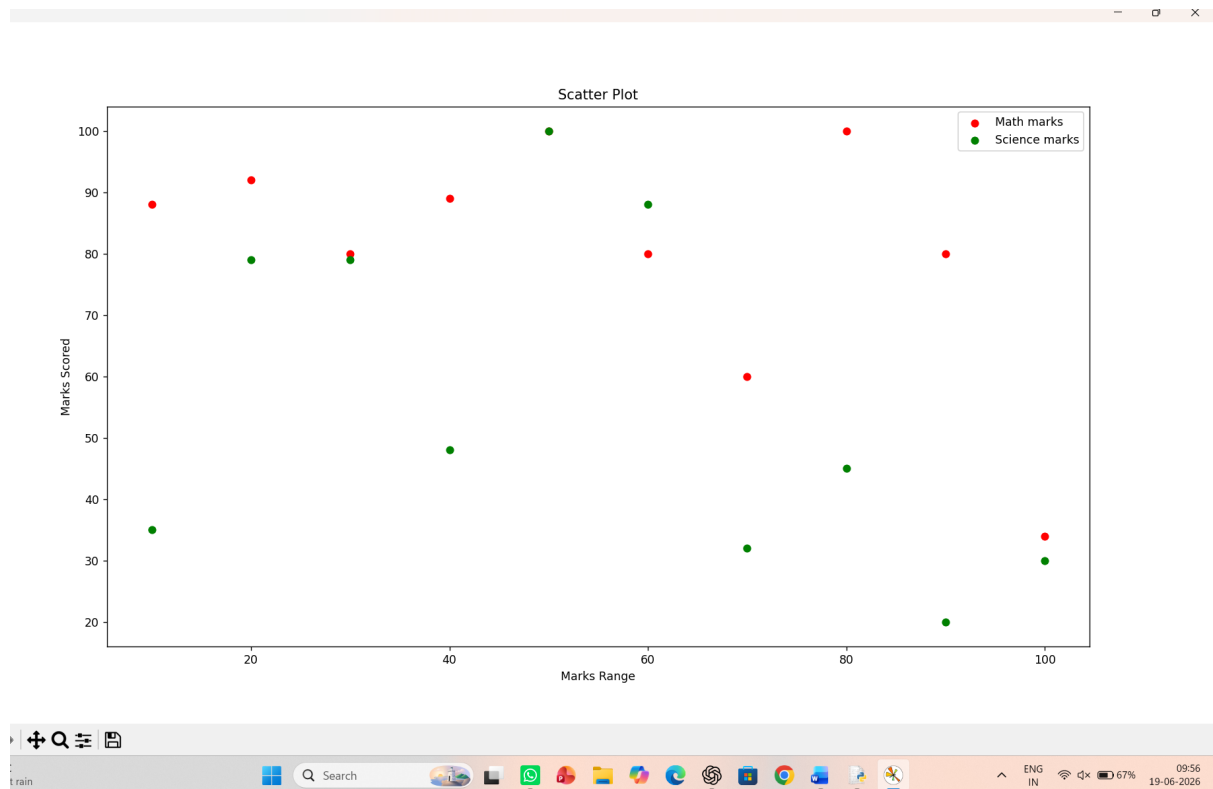
Algorithm

1. Import the Matplotlib library.
 2. Define the Mathematics marks, Science marks, and marks range data.
 3. Create a scatter plot for Mathematics marks using red color.
 4. Create another scatter plot for Science marks using green color.
 5. Add labels for the X-axis and Y-axis.
 6. Add a title and legend to the graph.
 7. Display the scatter plot.
-

Code

```
import matplotlib.pyplot as plt
math_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34]
science_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]
marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
plt.scatter(marks_range, math_marks,
            color='red',
            label='Math marks')
plt.scatter(marks_range, science_marks,
            color='green',
            label='Science marks')
plt.xlabel('Marks Range')
plt.ylabel('Marks Scored')
plt.title('Scatter Plot')
plt.legend()
plt.show()
```

Output



Result

Thus, the Python program to draw a **scatter plot comparing Mathematics and Science marks of 10 students** was successfully executed and the required graph was obtained.

Experiment No: 36

DATE:22/06/2026

Aim

To write a Python program to draw a **scatter plot for three different groups comparing weights and heights.**

Algorithm

1. Import NumPy and Matplotlib libraries.
 2. Generate random weight and height values for three groups.
 3. Plot the data of each group using different colors and markers.
 4. Add labels for the X-axis and Y-axis.
 5. Add a title and legend to the graph.
 6. Display the scatter plot.
-

Code

```
import matplotlib.pyplot as plt

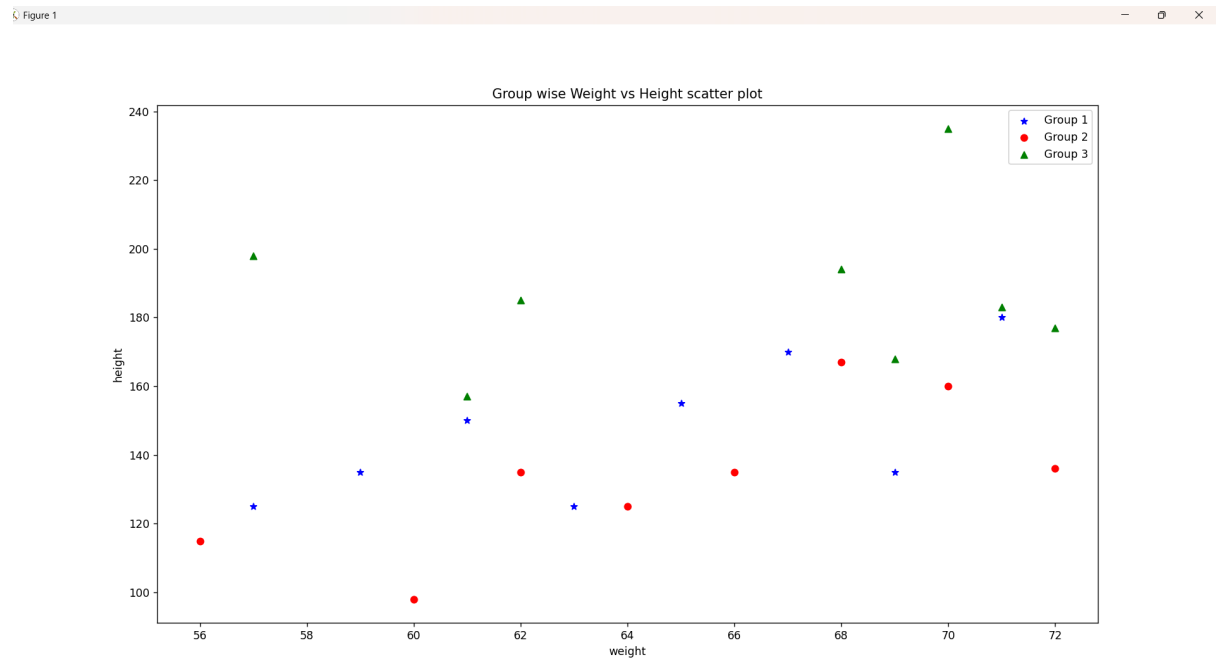
weight1 = [57, 59, 61, 63, 65, 67, 69, 71]
height1 = [125, 135, 150, 125, 155, 170, 135, 180]
weight2 = [56, 60, 62, 64, 66, 68, 70, 72]
height2 = [115, 98, 135, 125, 135, 167, 160, 136]
weight3 = [57, 61, 62, 68, 70, 71, 69, 72]
height3 = [198, 157, 185, 194, 235, 183, 168, 177]

plt.scatter(weight1, height1, color='blue', marker='*', label='Group 1')
plt.scatter(weight2, height2, color='red', marker='o', label='Group 2')
plt.scatter(weight3, height3, color='green', marker='^', label='Group 3')

plt.xlabel('weight')
plt.ylabel('height')
plt.title('Group wise Weight vs Height scatter plot')
plt.legend()
```

```
plt.show()
```

Output



Result

Thus, the Python program to draw a **scatter plot for three different groups comparing weights and heights** was successfully executed and the required graph was obtained.